

Paper: Zeng-Etal-An Empirical Evaluation of Columnar Storage Formats-VLDB23

Overview

The paper revisits two widely-used open-source columnar storage formats, Apache Parquet and Apache ORC, whose original design dates back to the early-2010s Hadoop ecosystem, and asks how well they serve modern hardware and workloads. The authors build a benchmark that stresses the formats under varying data distributions, encoding/compression choices, query selectivities, and even GPU-based processing. They then analyze the internal design decisions of the formats (encoding, metadata, filtering, nested types) and identify where they are well-aligned and where they are sub-optimal for today's analytics and ML workloads.

Key findings / take-aways

There is no clear “winner” between Parquet and ORC across all dimensions: for example, Parquet tends to produce slightly smaller files (thanks to aggressive dictionary encoding) and offers faster decoding in many cases, whereas ORC often yields better pruning/skip performance due to finer-grained zone maps.

Many real-world datasets have low “number of distinct values” (NDV) per column, making dictionary encoding very effective; thus, the integer encoding algorithm (after dictionary) becomes a critical performance dimension. Because modern storage and memory bandwidth have improved so much, the bottleneck is shifting from I/O to decoding/compute; accordingly, formats that favour faster decoding (rather than maximal compression) and lighter block compression often perform better.

However, for certain modern workloads such as ML training with thousands of features/projections, or vector embedding similarity search or GPU-based decoding both Parquet and ORC show inefficiencies (metadata parsing overheads, lack of fine parallelism, suboptimal codecs) and thus they are not ideal out of the box for these scenarios.

Description of the system discussed and how it was modified/extended

They built a benchmark framework (“OpenFormat” in their [GitHub repo](#)) which generates data based on real-world distributions, writes Parquet and ORC files under controlled parameter settings (row group/stripe sizes, encoding thresholds, compression codecs, metadata layouts), and then measures space usage, full scans, filtered scans, and GPU-based decoding. While they did not build a new storage format per se, they instrumented and extended reader/decoder implementations of Parquet/ORC (for example using Arrow C++ and cuDF for GPU tests) to isolate and profile internal components (encoding/decoding, metadata parsing, filtering).

Workloads/benchmarks used in evaluation

They created several synthetic workloads derived from real-world dataset statistics (named “core”, “bi”, “classic”, “geo”, “log”, “ml”) each representing varying column properties (NDV ratio, sortedness, value range, Zipf skew) and query selectivities; for each workload they generated a table (~1 million rows, 20 columns) and ran sequential scans and filtered queries on both formats.

References

1. Zeng, X., Hui, Y., Shen, J., Pavlo, A., McKinney, W., & Zhang, H. (2023). An empirical evaluation of columnar storage formats (arXiv:2304.05028) [Preprint]. arXiv. <https://arxiv.org/abs/2304.05028>
2. GitHub repository: embryo-labs. (2023). EvaluationOfColumnarFormats [Source code]. GitHub. <https://github.com/embryo-labs/EvaluationOfColumnarFormats>